Heterogeneity in Consumer Behavior: Micro Evidence and Macro Implications *

Preliminary and Incomplete

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Abstract

How well are Indian households insured against income shocks, and how does this vary across the income distribution? We estimate a panel unobserved components model of consumption and income dynamics using monthly data from India's Consumer Pyramids Household Survey, covering over 90,000 households from 2016-2019. We find three key results. First, household income is particularly volatile: transitory shocks are more persistent and the standard deviation of permanent shocks is substantially higher than in the US. Second, household consumption is highly sensitive to income changes: the quarterly marginal propensity to consume (MPC) out of a permanent income shock is 0.49, while the quarterly MPC out of a transitory income shock is 0.19. Third, MPCs decline in income deciles, educational attainment and age. A calibrated life-cycle model with incomplete asset markets reproduces consumption behavior in levels and in response to income shocks. Using this framework, we show that welfare is higher under tax systems with greater income tax progressivity and lower consumption tax rates in environments with high income risk.

Keywords: Income risk; consumption insurance; marginal propensities to consume; tax progressivity

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1 Introduction

The degree to which consumption changes in response to income fluctuations, called partial insurance in the terminology of Blundell, Pistaferri, and Preston (2008), determines both the welfare costs of income volatility and the effectiveness of fiscal interventions. However, little is known about consumption insurance in developing economies, where income volatility is higher and formal insurance mechanisms more limited.

India provides a compelling laboratory for studying consumption insurance. With 1.4 billion people, substantial income inequality, and limited access to formal financial markets, India represents the lived experience of a large fraction of the world's population. Indian households face high income volatility from sources ranging from monsoon variability to significant transitioning in and out of low-wage jobs (Donovan, Lu, and Schoellman, 2023), yet they also have access to extensive kinship networks and an expanding system of government transfers through programs like the Public Distribution System.

This paper uses high-frequency panel data from India's Consumer Pyramids Household Survey to provide the first comprehensive estimates of heterogeneous consumption insurance using a Kalman filter based maximum likelihood method. We estimate the semi-structural model of Blundell et al. (2008) using the quasi maximum likelihood methodology developed in Chatterjee, Morley, and Singh (2021). Our paper provides estimates of time-varying permanent and transitory income volatility, persistence of transitory income shocks, and most importantly, marginal propensities to consume out of permanent and transitory income shocks for a major emerging economy. Our analysis of over 90,000 households from 2016-2019 using quarterly data yields three main findings that differ markedly from developed country patterns.

First, the income shock environment in India is riskier than in developed countries. We estimate that permanent income shocks are substantially higher and transitory shocks more persistent than in the US data.

Second, in this riskier environment, Indian household consumption is highly sensitive to income shocks. The average transmission coefficient from permanent income shocks to consumption is 0.73, indicating that households smooth about one-fourth of permanent income changes. Combined with the average propensity to consume measures from the data, the estimated transmission coefficient implies an estimated quarterly marginal propensity to consume (MPC) of around 0.49, considerably higher than that in developed countries. Insurance against transitory shocks is also incomplete, with a transmission coefficient of 0.28 and quarterly MPC of 0.19.

Third, the MPC out of permanent income shocks follows a familiar declining pattern across the income distribution. The poorest households exhibit the highest MPC (around 0.7), which declines to around 0.2 for top income earners. Similar pattern of heterogeneity is observed over the life cycle, across education and caste groups.

We use a standard life cycle incomplete asset markets model (see Heathcote, Storesletten, and Violante (2009) for a survey) to explain these consumption patterns and conduct policy experiments. Households choose non-durable consumption and their asset position given disposable income and asset holdings. We also assume that households spend a constant fraction of disposable income on other consumption goods, which helps match the APC of non-durable consumption in the data. We calibrate the discount factor to match the transmission coefficient of a temporary shock.

Our calibrated model reproduces consumption behavior both in levels and in response to income shocks. It matches average non-durable consumption and financial asset holdings over the life cycle. Importantly, the model also matches the untargeted transmission coefficient to a permanent income shock. Moreover, it reproduces the declining MPC out of a permanent income shock across income deciles, measured using lagged permanent income in the model, and across age groups.

We use this model to investigate the welfare cost of idiosyncratic income risk and desired social policy. Welfare gains from eliminating income risk are substantial. Consumption-equivalent welfare would increase by 40% if both permanent and transitory risk were removed. Most of this gain comes from eliminating permanent risk, as intuition suggests. Specifically, removing permanent income risk increases welfare by 32.8%, while eliminating transitory risk raises welfare by nearly 8.5%.

The substantial welfare gains from eliminating income risk suggest that stronger social insurance can increase welfare. We examine the welfare implications of policy by varying either income tax progressivity or the consumption tax rate. In these experiments, we maintain net-revenue neutrality by adjusting either the consumption tax rate or the average income tax rate. We find that welfare increases with higher income tax progressivity and lower consumption tax rates. When consumption tax rates are bounded below by zero, welfare is maximized at a zero consumption tax rate and an income tax progressivity of 25%.[1]

These findings have important policy implications, particularly for emerging economies where indirect taxes are the main source of government revenue (Bachas, Jensen, and Gadenne, 2024) and income risk is high. In such settings, our results suggest that welfare is higher under

[1] In this experiment, the average income tax rate is fixed at its calibrated value of -8%.

tax systems with greater income tax progressivity and lower consumption taxes.

In addition, as Kaplan and Violante (2010) recommend, empirical estimates generated by models such as ours can serve as yardsticks for quantitative macroeconomic models, particularly the heterogeneous-agent, incomplete markets models that have become the workhorse of modern macroeconomics. We hope that the results generated in this paper can therefore provide important targets for the quantitative macroeconomic literature on India and other emerging countries.

Our work contributes to three strands of literature. We extend the consumption insurance literature pioneered by Blundell et al. (2008) to a major developing economy, demonstrating how institutional differences shape household responses to income shocks. We contribute to the fiscal policy literature by providing micro-foundations for heterogeneous MPC patterns that affect transfer multipliers (Kaplan and Violante, 2014; Orchard, Ramey, and Wieland, 2025). Finally, we contribute to the literature on policy design by quantifying welfare implications of income and consumption tax progressivity (Wu and Krueger, 2021).

The remainder of the paper proceeds as follows. Section 2 describes the Consumer Pyramids Household Survey data. Section 3 presents our empirical methodology and main results on income processes and consumption insurance. Section 4 develops and calibrates our life-cycle model. Section 5 presents our policy counterfactuals. In this preliminary version of the paper, we discuss proposed counterfactual experiments in the concluding Section 6.

2 Data

We use longitudinal data from India's Consumer Pyramids Household Survey (CPHS), which is conducted by the Center for Monitoring Indian Economy (CMIE). The survey includes a representative sample of over 174,000 households, making it the world's largest household panel survey (University of Pennsylvania Library, 2025). It is a continuous survey that began in 2014 where each household is surveyed at an interval of four months. For example, if a household is first surveyed in February of a particular year, it is surveyed again in June and October of that year.[2]

The CPHS sample size and frequency offers several advantages that make it suitable to address the questions that we pose. The frequency of data collection makes it possible to observe income shocks and consumption responses that would be missed at annual or lower frequencies. Higher-frequency data can be particularly valuable for distinguishing between transitory

[2] For additional details, see Vyas (2020).

and permanent income shocks, a distinction that is particularly relevant in contexts where seasonal employment and informal work arrangements may also contribute to income volatility. The higher frequency also likely reduces recall bias, as respondents report more recent income and consumption rather than trying to recall numbers from further in the past.

We merge household and individual modules of CPHS to construct total earnings and non-durable consumption at the household level, employment status and years of education of the survey respondent, and data on household demographics. All nominal values are converted to real terms using state-level CPI deflators.

2.1 Sample Selection

Our sample selection process begins with the universe of households in the CPHS data. Starting from the raw merged dataset of 231,371 households, we restrict attention to married households and limit analysis to household heads aged 25-65 years to concentrate on working-age households. We focus on 2016-2019 to ensure a consistent macroeconomic environment and trim the top and bottom 1% of income growth observations to address outliers. Most importantly, we require households to have non-missing income and consumption data for at least 12 of the 16 quarters. This process yields our final analysis sample of 90,817 households. Sample restrictions are described in Table A.2.

2.2 Descriptive Statistics

Table 1 presents comprehensive descriptive statistics for our estimation sample. The data reveal substantial heterogeneity across demographic, income, and consumption dimensions that make the sample well-suited for studying consumption insurance patterns.

2.2.1 Demographic Characteristics

Our analysis sample contains households with a median size of 4 members and typically includes 1 child under age 18. The median age of household heads is 45 years, reflecting our focus on working-age populations. Educational attainment varies considerably: 45% of household heads have 0-7 years of schooling, 37% completed 8-12 years, and 18% have more than 12 years of education.

Geographic distribution shows that 34% of households reside in urban areas, while 6% live in major metropolitan centers ("big cities"). Notably, 33% of households are engaged in agricul-

Table 1: Descriptive Statistics - CMIE Consumer Pyramids Household Survey

	Our Sample		Full Sample	
	Median	Standard Deviation	Median	Standard Deviation
Panel A: Demographic Variables				
Household size	4.00	1.48	5.00	2.03
Number of children 18 and under	1.00	1.29	1.00	1.20
Age of household head	45.00	9.48	26.00	46.18
Educational Attainment (%)				
0-7 Years	44.87	49.74	55.33	49.71
8-12 Years	36.62	48.18	35.42	47.83
> 12 Years	18.52	38.85	9.25	28.97
Urban household (%)	34.08	47.40	33.16	47.08
Agriculture (%)	33.58	47.23	30.55	46.06
Panel B: Income Measures				
Monthly labor earnings	10666.67	10926.38	8000.00	11093.87
Log monthly labor earnings	9.33	0.65	9.27	0.73
Growth rate of labor earnings (quarterly, %)	0.00	4382.49	-0.36	5118.78
Income percentiles:				
10th percentile	5233.33		0.00	
50th percentile (median)	11733.33		10333.33	
90th percentile	29690.00		29760.00	
Income sources (% of total):				
Labor earnings	99.24	21.15	98.26	36.23
Business income	0.00	15.22	0.00	24.91
Transfer income	0.09	8.63	0.20	16.58
Other income	99.75	17.35	99.49	28.91
Panel C: Consumption Measures				
Monthly nondurable consumption	6303.33	3061.31	6803.33	4837.98
Log monthly nondurable consumption	8.77	0.42	8.91	0.53
Consumption-to-income ratio	62.12	722.61	61.34	1084.99
Growth rate, nondurable consumption	-0.11	24.76	-0.28	35.26
Consumption shares (% of total):				
Food	59.37	9.98	58.58	10.10
Housing	0.00	7.36	0.00	6.96
Transportation	2.63	2.55	2.67	2.81
Education	1.79	9.16	0.95	9.12
Healthcare	2.20	8.78	2.32	18.08
Other	29.58	17.68	30.57	23.65

Notes: This table presents descriptive statistics for our analysis sample from the CMIE Consumer Pyramids Household Survey covering the period January 2016-December 2019. The sample includes households with heads aged 25-65 years with at least 12 quarters of non-missing income and consumption data. All monetary values are nominal Indian Rupees.

ture, reflecting the continued importance of this sector in the Indian economy. The unemployment rate in our sample is 5.5%, consistent with the focus on households with regular income reporting.

2.2.2 Income Patterns

The income distribution exhibits substantial inequality and volatility, characteristic of developing economies. Median monthly real (in 2011 prices) labor earnings are ₹ 10,666 (approximately $120 at 2026 exchange rates) with a standard deviation of ₹ 10,926, indicating considerable dispersion. The income distribution is highly right-skewed: the 10th percentile (₹ 5,233) represents subsistence-level earnings, the median captures lower-middle-class households, and the 90th percentile (₹ 29,690) reflects relatively affluent households by Indian standards.

Income volatility is substantial and underscores the importance of understanding how households respond to income fluctuations. The predominance of labor earnings is striking—for the median household, wage income comprises 99% of total income, with minimal reliance on business income (0%), transfer payments (0.09%), or other sources.

2.2.3 Consumption Behavior

Monthly nondurable consumption patterns reveal both the economic constraints and smoothing behavior of Indian households. The median household spends ₹ 6,303 (real terms in 2011 prices) monthly on nondurable goods, with a standard deviation of ₹ 3,061 that is proportionally much smaller than income dispersion. This suggests some degree of consumption smoothing even in the raw data.

The consumption-to-income ratio has a median of approximately 62%, indicating that typical households save about 38% of their income. However, the large standard deviation (722%) reflects substantial heterogeneity in saving behavior across the income distribution. The quarterly growth rate of nondurable consumption shows a median near zero (-0.11%) with a standard deviation of 25%, markedly lower volatility than income growth, providing preliminary evidence of consumption smoothing.

The composition of consumption reflects the relatively low income levels and spending priorities typical of developing economies. Food represents the largest expenditure category at 59% of nondurable consumption for the median household. Housing costs show a median share of zero, likely reflecting widespread home ownership or residence in family compounds without explicit rental payments.[3] Other significant categories include transportation (2.7%), healthcare (2.2%), and education (1.8%), with the remainder (29%) comprising various other nondurable goods and services.

[3] Badarinza, Balasubramaniam, and Ramadorai (2016) document that real estate ownership in India is substantially higher than in countries such as the US, the UK, and China.

2.2.4 Sample Characteristics for Consumption Insurance Analysis

Several features of our sample make it well-suited for analyzing consumption insurance. First, the substantial income inequality—spanning from near-subsistence to relatively affluent households—enables examination of heterogeneous consumption responses across the income distribution. Second, the high income volatility provides ample variation in permanent and transitory shocks necessary for identification. Third, the quarterly frequency and multi-year panel structure allow separation of permanent from transitory income components.

The predominance of labor income simplifies the analysis by focusing attention on employment-related shocks rather than complex portfolio responses. Finally, the substantial variation in consumption-income ratios across households suggests meaningful differences in saving behavior and liquidity constraints that may drive heterogeneous consumption responses to income shocks.

3 Empirical Model & Results

We follow Blundell et al. (2008), contextualized for salient demographic characteristics of India, to obtain residual income and consumption from the first stage regression. In particular, we include deterministic controls such as year and state fixed effects, age, education categories, number of household members, number of kids, big city and state interacted with year, caste and religion, employment status. Residual log income and log consumption can be characterized as:

$$y_{i,t} = \tau_{i,t} + \epsilon_{i,t} + \theta\epsilon_{i,t-1}, \qquad \epsilon_{i,t} \sim i.i.d.(0,\sigma_\epsilon^2), \tag{3.1}$$

$$c_{i,t} = \gamma_\eta\tau_{i,t} + \kappa_{i,t} + \upsilon_{i,t}, \qquad \upsilon_{i,t} \sim i.i.d.(0,\sigma_\upsilon^2), \tag{3.2}$$

Here $\tau_{i,t}$ is a common stochastic trend for income and consumption ("permanent income"), $\epsilon_{i,t}$ is a transitory income shock with moving-average parameter $|\theta| < 1$, $\kappa_{i,t}$ is an additional stochastic trend for consumption, and $\upsilon_{i,t}$ is a transitory consumption shock. The trends are specified as random walks:

$$\tau_{i,t} = \tau_{i,t-1} + \eta_{i,t}, \qquad \eta_{i,t} \sim i.i.d.(0,\sigma_\eta^2), \tag{3.3}$$

$$\kappa_{i,t} = \kappa_{i,t-1} + \gamma_\epsilon\epsilon_{i,t} + u_{i,t}, \qquad u_{i,t} \sim i.i.d.(0,\sigma_u^2), \tag{3.4}$$

We allow the variance of permanent and transitory income shocks, σ_η^2 and σ_ϵ^2, to vary by quarter. We estimate all the parameters of the semi-structural model, $\theta, \gamma_\eta, \gamma_\epsilon, \sigma_u, \sigma_v, \sigma_{\eta,t}, \sigma_{\epsilon,t}$, following quasi maximum likelihood (QMLE) based approach of Chatterjee et al. (2021). We present a state space representation of the Blundell et al. (2008) model, following Chatterjee et al. (2021), in the online appendix.[4]

Sun (2024) has found important differences in household consumption insurance for India using GMM estimation method of Blundell et al. (2008). Motivated by these findings, we report results both for the overall population and also provide comparisons for various subgroups. We also report estimates for the US from the literature (Blundell et al. (2008) and Chatterjee et al. (2021)) to offer a baseline for comparison. However, note that the estimates based on PSID data reported in both Blundell et al. (2008) and Chatterjee et al. (2021) use annual data, while we use quarterly data in our estimation. Auclert (2019) and Hong (2023) provide careful detail about how to compare the annual versus quarterly estimates of MPC. In the current version of the paper, we have not yet completed this comparison. So, instead of providing a quantitative comparison, we provide a qualitative discussion about how our estimates differ compared to the US.

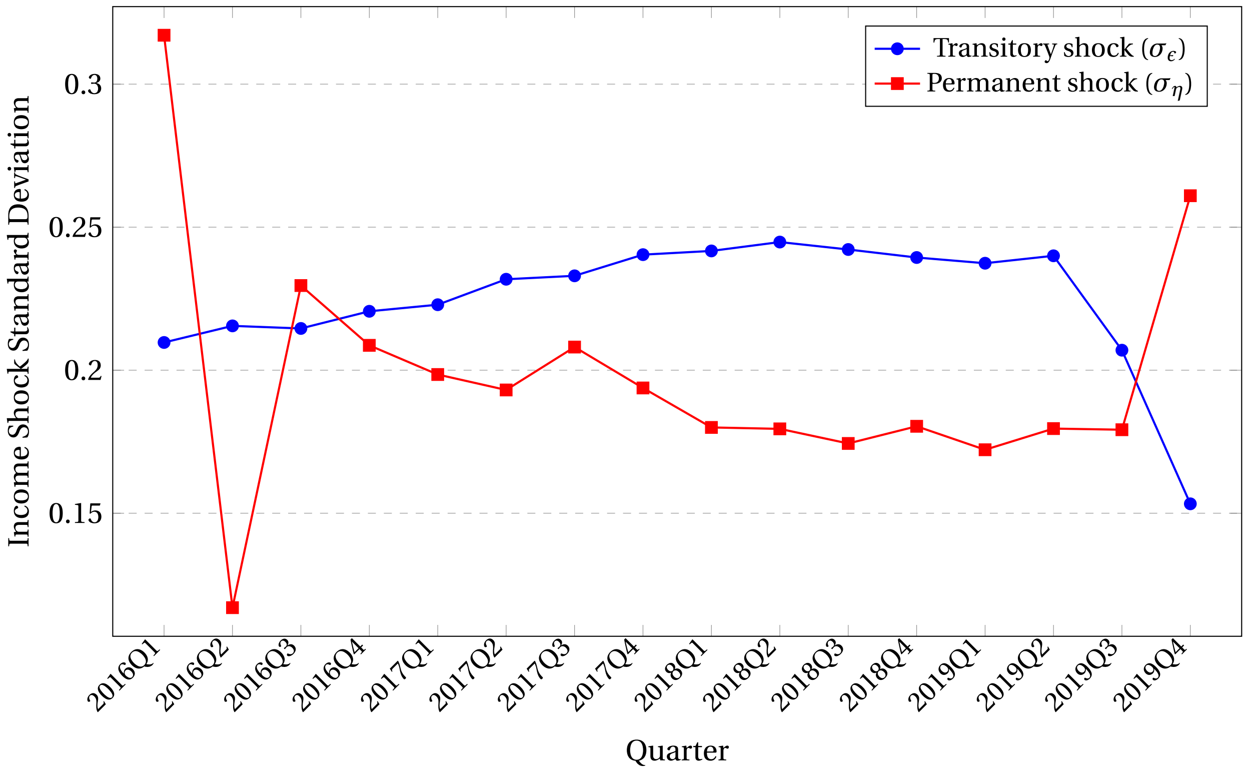
How volatile and persistent are income shocks in India? We begin first by looking at the standard deviation (SD) of the permanent and transitory shocks over time. Figure 1 displays the quarterly estimates of $\sigma_{\eta,t}$ and $\sigma_{\epsilon,t}$. In general, our estimates reveal that the permanent income shock in India is more volatile and the transitory shock is more persistent than in the US. Consistent with the literature, the transitory shock is more volatile than the permanent shock to income.

What is the estimated transmission coefficient of permanent income shocks to consumption, γ_η that governs household consumption insurance against permanent shocks? Table 2 reports the estimates for the full sample and compares against the estimates for the US using the PSID data by applying the same semi-structural model of (Blundell et al., 2008) and QMLE method of Chatterjee et al. (2021). The estimated transmission coefficient out of permanent income shocks is 0.73 for the full sample; consumption changes less than one-for-one with the income shock implying partial self insurance. The transmission coefficient is noticeably higher, and the resulting degree of self insurance lower than that of the US.[5] The estimated transmission coefficient of transitory shocks is also much larger in India revealing low degree of self

[4]The estimation method is described as quasi maximum likelihood because in describing the unobserved components model in equations 3.1, 3.2, 3.3 and 3.4 we do not make any distributional assumption, while in writing the likelihood function based on the state space model we need to assume Normal distributions.

[5]A transmission coefficient (γ_η) of permanent income shocks on consumption implies self insurance of 1-γ_η. As shown in Table 2, in India using CMIE data the estimated self-insurance is 0.27.

Figure 1: Standard Deviation of Income Shocks



insurance against volatile and persistent transitory income shocks in Indian data.

The marginal propensity to consume (MPC) out of an income shock equals the transmission coefficient to that shock multiplied by the average propensity to consume (APC). For India, quarterly MPC out of a permanent shock is 0.49, while the quarterly MPC out of a transitory shock is 0.19. Auclert (2019) and Hong (2023) show that annual MPC is equal to $1 - (1 - \text{quarterly MPC})^4$. This implies that for India, annual MPC out of a permanent and transitory income shock are 0.93 and 0.57, respectively.

Our primary interest is to assess the pattern of transmission coefficient, APC and ultimately MPC across the income distribution to gauge the relationship between income and the degree of household consumption insurance. In Figure 2 we report the estimated transmission coefficients, while Figure 3 also reports the APC measures and the implied MPC out of the permanent shock by income decile. Both the transmission coefficient and the APC decline monotonically across the income distribution. As a result, the implied MPCs decline with income.

To further corroborate this finding, we use the richness of our data and the precision of our estimation method, to estimate the entire model for a wide range of subgroups. Consistent with

Table 2: Estimates of Income Risk, Persistence & Transmission Coefficients

Parameter	CMIE	PSID
Average standard deviation of permanent shock (σ_η)	0.198	.1
Average standard deviation of temporary shock (σ_ϵ)	0.225	.245
Persistence (θ)	0.2953	0.16
	(0.0042)	(0.02)
Transmission coefficient to permanent shock (γ_η)	0.7308	0.46
	(0.0038)	(0.04)
Transmission coefficient to transitory shock (γ_ϵ)	0.2784	0.03
	(0.0020)	(0.02)

Figure 2: Transmission coefficient across Income Distribution

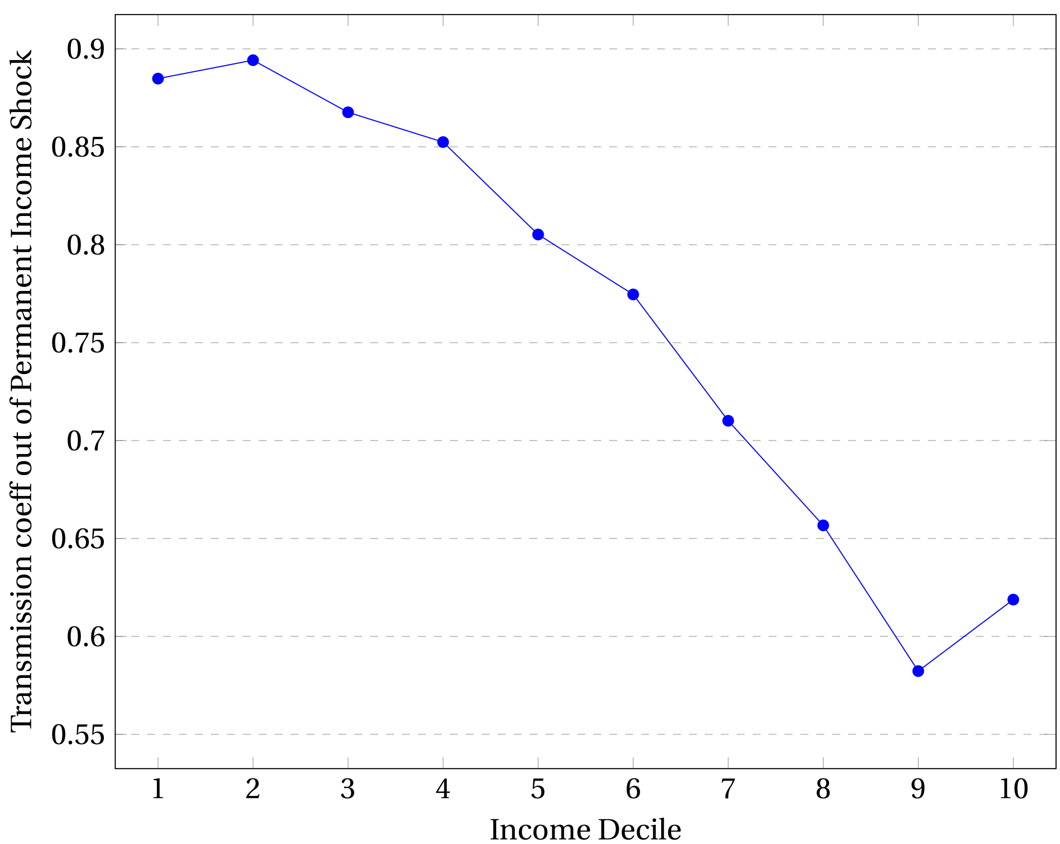
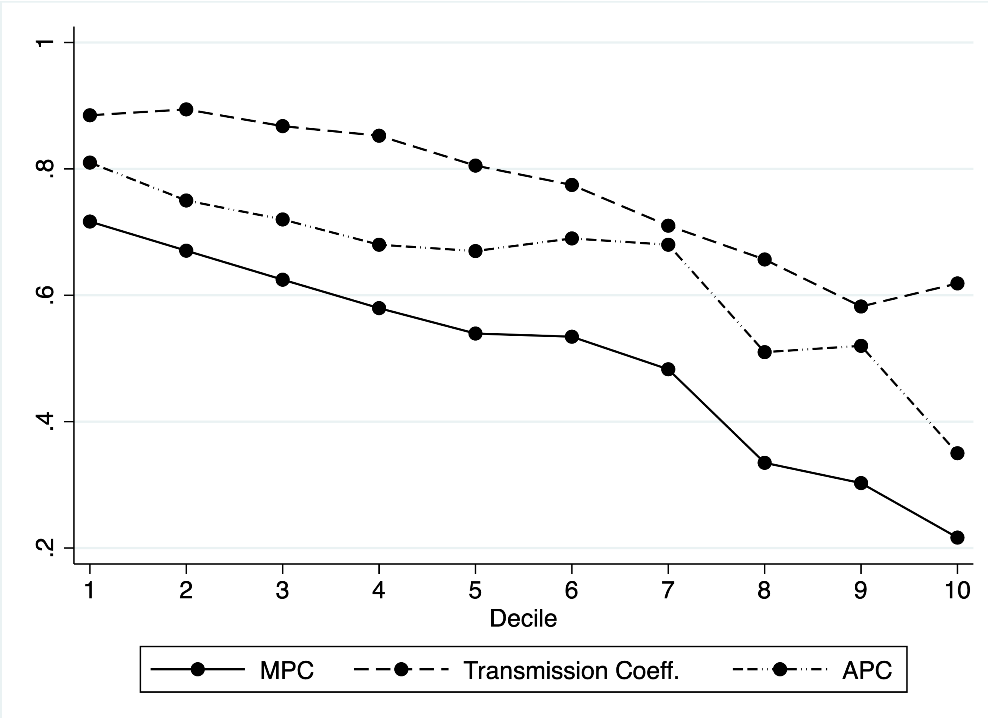


Figure 3: Transmission Coefficient, APC and MPC across the Income Distribution



this pattern, we find that in Table 3 that transmission coefficient to a permanent income shock decline in educational attainment and are higher for the historically disadvantaged reserved (scheduled caste and scheduled tribe) caste categories.

In Figure 4 we report the estimated transmission coefficients by different age bins, while Figure 5 also reports the APC and the implied MPC out of the permanent shock by the age bins. Younger households tend to be asset-poor and have lower earnings. Thus, consistent with the pattern in Figures 2 and 3, transmission coefficients and MPC both decline with age in Figures 4 and 5.

Our analysis confirms that (a) permanent income shock is far more volatile in India, while transitory income shocks are more persistent than US; (b) the average transmission coefficient of both permanent and transitory income shocks to consumption are substantially higher in India implying a low degree of self insurance; and (c) households with lower income, measured either by deciles of current income or various socioeconomic proxies, exhibit a higher consumption coefficient and higher average propensity to consume implying high MPCs. These empirical patterns have rich implications for public policy: both policies that aim to directly address volatility of income, and various income and consumption tax policies. In the next Sec-

Figure 4: Life-Cycle Patterns in Consumption Responses

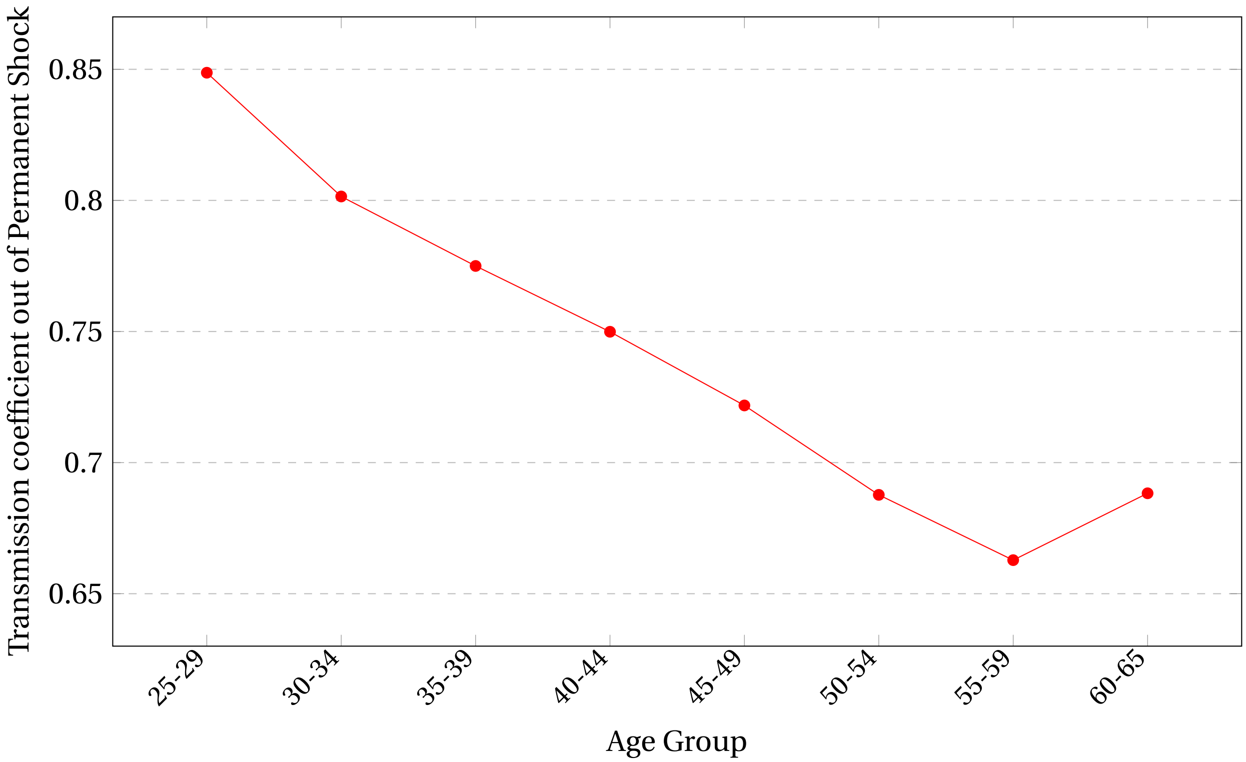
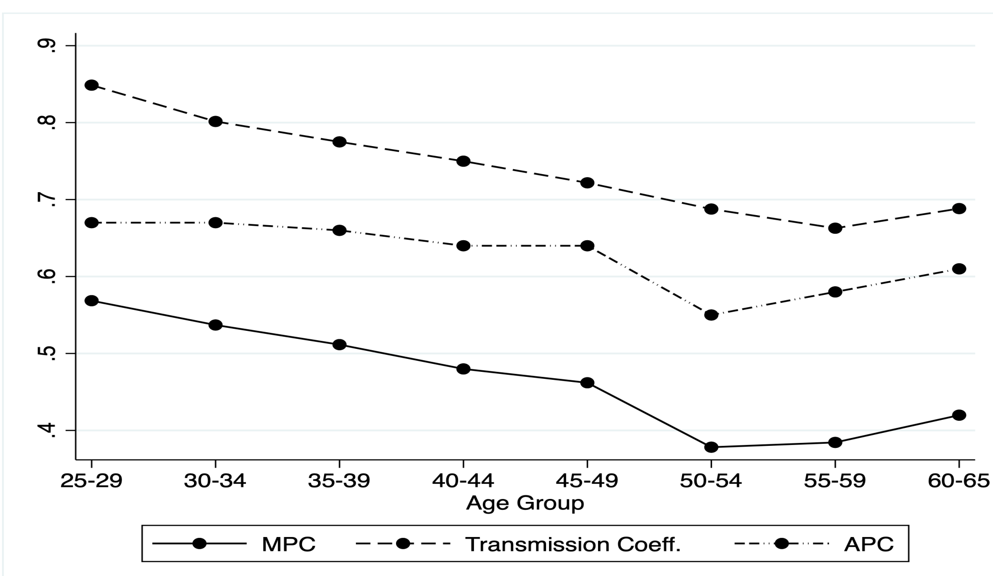


Figure 5: Transmission Coefficient, APC and MPC across the Life Cycle



tion, we build a standard incomplete markets model with idiosyncratic risk, calibrated to Indian data, to evaluate welfare implications of a variety of policy paradigms.

Table 3: Education and Social Groups: Consumption Insurance

Education Level	Permanent	Transitory
Primary	0.8086	0.2900
Secondary	0.7234	0.2730
College	0.6267	0.2661
Social Group	**Permanent**	**Transitory**
Non-reserved	0.7144	0.2719
Reserved	0.7713	0.2938

4 Life-Cycle Model to Rationalize Empirical Findings

Our empirical analysis shows that households display limited insurance against income shocks, particularly permanent income shocks. To rationalize these findings, we use a standard life-cycle model with idiosyncratic income risk and incomplete asset markets (Deaton, 1991; Carroll, 1992; Kaplan and Violante, 2010).[6]

4.1 Model Environment

The economy consists of a continuum of households of measure one, indexed by i. Time is discrete; a period corresponds to a quarter in the model. Households work from age 1 to R periods of life. They retire at age $R+1$ and live until age T. To focus on idiosyncratic income risk, we assume there is no aggregate uncertainty.

Households maximize time-separable expected utility derived from non-durable consumption C_t. Expected utility is described as:

$$E_0 \sum_{t=0}^{T} \beta^t \frac{C_t^{1-\sigma}}{1-\sigma} \tag{4.1}$$

where $\beta \in (0,1)$ is the discount factor and $\sigma > 0$ is the coefficient of relative risk aversion.

[6]We abstract from additional sources of uncertainty during retirement such as survival risk (Yaari, 1965) or health risk (De Nardi, French, and Jones, 2010).

Households receive pre-tax labor income Y_{it}, which during working years comprises of three components: (i) a deterministic component (Γ_t) that is common to all households and varies by age, (ii) a permanent income component $\left(\tau_{i,t}\right)$ that follows a random walk process, and (iii) temporary shocks with some persistence. Household income process follows the same process as in the empirical setup in Section 3.

Following Benabou (2002), household post-tax income is a concave function of pre-tax labor income. We model government transfers as government providing an income floor $Y_{\min}$. Therefore, household disposable income is given by:

$$Y_{i,t}^{p} = \min\left\{\chi_y\left(Y_{i,t}\right)^{1-\psi}, Y_{\min}\right\} \tag{4.2}$$

where $Y_{i,t}^{p}$ is disposable labor income, ψ determines the marginal tax rate and χ_y is an income-scaling parameter affecting the average income tax.

Households save in a risk-free financial asset $\left(A_{i,t+1}\right)$ that pays a constant after-tax return of $1+r$. We assume a no-borrowing constraint such that $A_{i,t+1} \geq 0$. Additionally, we assume that households spend an exogenous fraction, $\bar{d}$, of their disposable income on durable consumption goods and non-financial assets. This allows us to match the average propensity to consume non-durable goods out of income (APC) in the data.[7]

Household budget constraint is:

$$\left(1+\chi_c\right)C_{i,t} + A_{i,t+1} = (1+r)A_{i,t} + Y_{i,t}^{p} \times \left(1-\bar{d}\right) \tag{4.3}$$

where χ_c is proportional consumption tax.

A household's optimization problem during working periods can be written in recursive formulation as:

$$\begin{aligned}
V(A,\zeta,\tau,t) &= \max_{A',C} \frac{C^{1-\sigma}}{1-\sigma} + \beta\,\mathbf{E}_{\zeta',\varepsilon'} V\left(A',\zeta',\tau',t+1\right) \\
&\text{subject to:} \\
\left(1+\chi_c\right)C + A' &= (1+r)A + Y^{p} \times \left(1-\bar{d}\right) \\
A' &\geq 0
\end{aligned}$$

Lastly, retired households receive pension benefits that are proportional to their permanent

[7]To rationalize this formulation, $\bar{d}$ can be interpreted as the share of income spent on durables when households derive utility from durables and non-durables consumption using a Cobb-Douglas utility function.

income level in the last working period. Their pre-tax income is:

$$Y_{i,t} = b \times \exp\left(\Gamma_R + \tau_{i,R}\right) \quad \forall t \in \{R+1, \ldots, T\}$$

where b is the replacement rate. Their optimization problem during retirement periods is:

$$\begin{aligned}
V(A, \tau, t) &= \max_{A', C} \frac{C^{1-\sigma}}{1-\sigma} + \beta\, \mathbf{1}_{t+1 \leq T}\, V\left(A', \tau, t+1\right) \\
\text{subject to:} & \\
\left(1 + \chi_c\right) C + A' &= (1+r) A + Y^p \times \left(1 - \bar{d}\right) \\
A' &\geq 0
\end{aligned}$$

The transmission coefficient out of a permanent income shock in the model is:

$$\gamma_t^{\eta} = \frac{\mathrm{Cov}\left(\Delta \log C_{it}, \eta_{it}\right)}{\mathrm{Var}\left(\eta_{it}\right)} \tag{4.4}$$

The MPC out of a permanent income shock equals the transmission coefficient multiplied by the average propensity to consume (APC):

$$\mathrm{MPC}_t^{\eta} = \frac{\mathrm{Cov}\left(\Delta \log C_{it}, \eta_{it}\right)}{\mathrm{Var}\left(\eta_{it}\right)} \times \frac{\sum_i C_{it}/N}{\sum_i Y_{it}/N} = \gamma_t^{\eta} \times \mathrm{APC}_t \tag{4.5}$$

Similarly, the transmission coefficient and MPC of a transitory income shock is defined as:

$$\begin{aligned}
\gamma_t^{\epsilon} &= \frac{\mathrm{Cov}\left(\Delta \log C_{it}, \epsilon_{it}\right)}{\mathrm{Var}\left(\eta_{it}\right)} && (4.6) \\
\mathrm{MPC}_t^{\epsilon} &= \gamma_t^{\epsilon} \times \mathrm{APC}_t && (4.7)
\end{aligned}$$

In the model, consumption tax and positive net income tax contribute to government revenue. On the other hand, government expenditure comprises of negative net income tax (transfers) and pension benefits during retirement periods. Therefore, net revenue for the government is:

$$\chi_c \sum_{t=1}^{T} \sum_i C_{i,t} + \sum_{t=1}^{R} \sum_i \left(Y_{i,t} - Y_{i,t}^P\right) - \sum_{t=R+1}^{T} \sum_i Y_{i,t}^P \tag{4.8}$$

We do not impose that the government budget clears, but we will ensure net revenue neutrality across counterfactual tax experiments.

4.2 Calibration and Life cycle patterns

We parametrize the model to reproduce key features of the Indian economy.

Demographics and Initial Asset Holdings: Households are born at age 25 ($t = 1$), retire at age $R = 65$ ($t = 164$), and live until age $T = 71$ ($t = 185$). We use data on financial assets at age 24 ($t = 0$) from the All India Debt and Investment Survey (AIDIS) in 2019 to calibrate asset holdings at the start of the life-cycle. We measure financial assets in the data as the sum of cash-in-hand, amount in current and savings bank account and other financial institutions, market value of stocks and debentures, and cash value of debt that the household will receive.

Preferences: We calibrate β to match the transmission coefficient into consumption of transitory income shocks. The quarterly value of β is 0.902, which suggests our results are not driven by a small discount factor. We calibrate $\bar{d}$ at 31% to match the empirically observed non-durable APC of 47.3%. The value of risk aversion σ is set at 2, a standard value in the literature.

Real Interest rate: We set the quarterly risk-free interest rate (r) to 0.27%. This value corresponds to an annual real interest rate of 1.1%. Risk-free interest rate is defined as nominal long-term interest rate minus expected inflation, measured as the average of current inflation and lagged inflation of the previous three quarters.

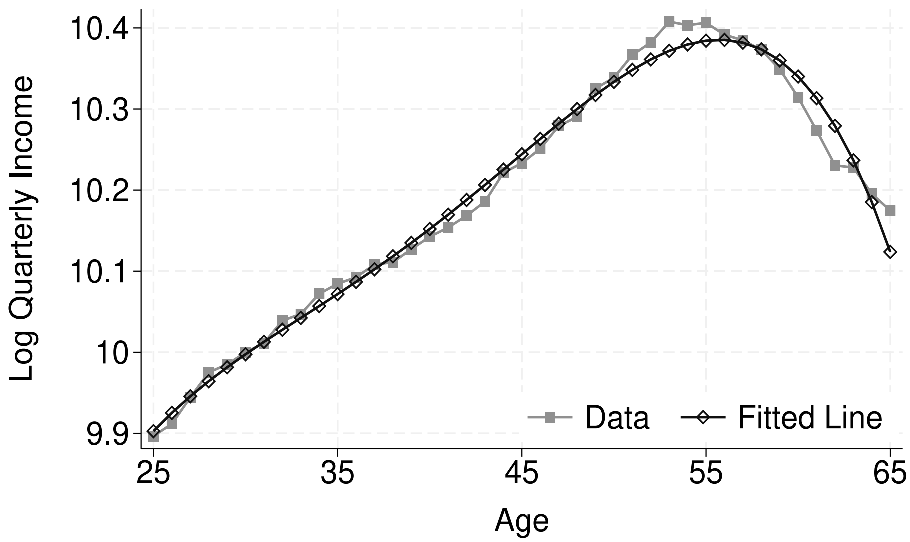
Income Process: The parameter values of $\sigma_\epsilon = 0.2246; \theta = 0.2953; \sigma_\eta = 0.198$ are taken from the estimated income process. The cross-sectional variance at age 24 ($t = 0$) of 0.11 is used to calibrate the initial draw of the permanent income level.

We use a fourth-order polynomial in age on log earnings to calibrate the deterministic life cycle earnings component. Figure 6 shows the estimated deterministic life-cycle earnings profile matches the data well.

Pension benefits are assumed to be 60% of the permanent income level in the last working period, which is a typical value in the literature.

Tax system and government transfers: We replicate the Indian tax system by setting the tax progressivity parameter ψ to 0.1550 and the income-shift parameter χ to 1.081, following Chakrabarti, Mishra, and Mohaghegh (2024). This implies an average income tax rate of $1-\chi_y = -0.081$. The value of proportional consumption tax χ_c is set at 16.37% to match the average of the consumption tax revenue relative to aggregate household consumption between 2016 and 2019 for India. Data on consumption tax revenue comes from Mansour, Verhoeven, Sawadogo, and Tan (2025).

Figure 6: Deterministic Income along the life cycle



Model Fit and Life cycle patterns: We solve for optimal policy functions using the endogenous grid method of Carroll (2006). We simulate a panel of 50,000 households from ages 25 to 85. The model is solved using 101 exponentially-spaced grid points for assets. We discretize the grid of the permanent and temporary component using a Markov chain with 25 and 15 equally spaced points.

Table 4 shows the fit of the model with regards to the moments targeted in the calibration exercise and some untargeted moments. We match the targeted transmission coefficient to a transitory shock (γ_{ϵ}) and the APC quite well. The model reproduces several untargeted moments in the data, such as the transmission coefficient to a permanent shock (γ_{η}) and the log of 1+financial wealth. This suggests that parameters such as the minimum government income and pension benefits, have been parametrized well, otherwise, we would not have been able to reproduce these moments in the model.

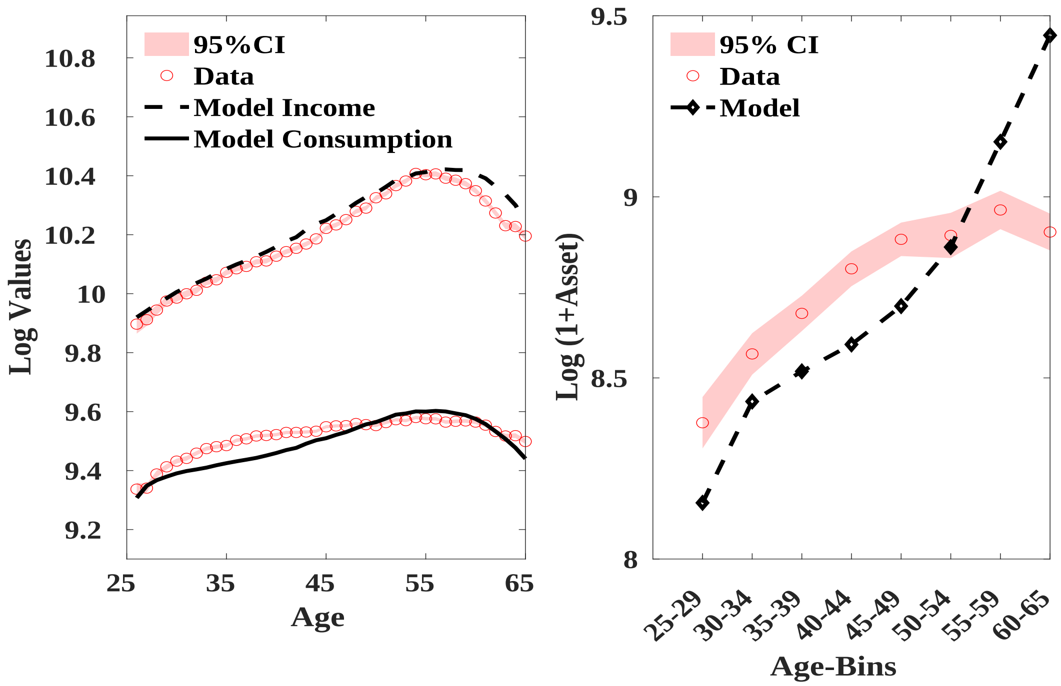
We compare the model-implied and empirical life-cycle profiles of means of income, consumption and, assets in Figure 7. The left-hand panel of the figure shows the life cycle profiles of logarithm of consumption and income. We targeted the average APC, however, the model reproduces the untargeted consumption profile over the life cycle in the data. Moreover, the model matches the low level of liquid wealth of Indian households (average of Rupees 6,567) and the broad pattern over the life cycle. It slightly under-predicts wealth during initial working periods and over predicts during close to retirement period. The over prediction during retire-

Table 4: Model Fit

Moment	Data Value	Model Value
A: Targeted		
Transmission coefficient to temporary shock	0.2784	0.2756
APC	47.3%	47.6%
B: Untargeted		
Transmission coefficient to permanent shock	0.7308	0.7362
Log (1+Wealth)	8.79	8.83

ment period might be due to the prevalence of multi-generational household structure in India, leading to lower asset accumulation in the data.

Figure 7: Life cycle profiles for means



Note: Data for mean of logarithm of income and consumption is from CMIE 2016 to 2019. Financial wealth data is from AIDIS 2019.

The model underperforms relative to matching consumption and inequality in the data over, as shown in Appendix Figure B.1. In the data, the standard deviation of residualized income is at most 41% higher between the highest standard deviation across age and at age 25.

However, it rises by more than 450% in the model. This implies that consumption inequality also rises by more than 400% in the model, while in the data it rises by at most 10-15% relative to age 25. Since inequality is quite high in this model, welfare gains from removing income risk will also be quite high, as we discuss below. Additionally, changes in tax structure will impact welfare more strongly than if inequality was lower.

MPC and transmission coefficients by lagged permanent income and age in the model: The top panel of Figure 8a shows the MPC and transmission coefficients out of a permanent and temporary income shock and APC by quintiles of lagged permanent income, whereas the bottom panel Figure 8b shows them along the life cycle. The MPC and transmission coefficients by quintiles of lagged permanent income level are averaged over the life cycle. The transmission coefficient out of permanent or transitory income shocks are fairly flat or slightly decreasing by lags of permanent income, except for the transmission coefficient of the lowest permanent income quintile which is lower than other quintiles. PIH households with higher permanent income consume less, which implies that MPC out of income shocks are decreasing in permanent income quintiles. The falling MPC for higher income quintiles is broadly in line with the observed empirical patterns.

Similarly, APC is falling over the life cycle as households consume less out of their income as they get older. Furthermore, the pass-through of an income shock into consumption dampens as households accumulate more assets over their life cycle. Taken together, the MPC out of income shocks are falling with age, which is also in line with the data.

5 Results from model counterfactuals

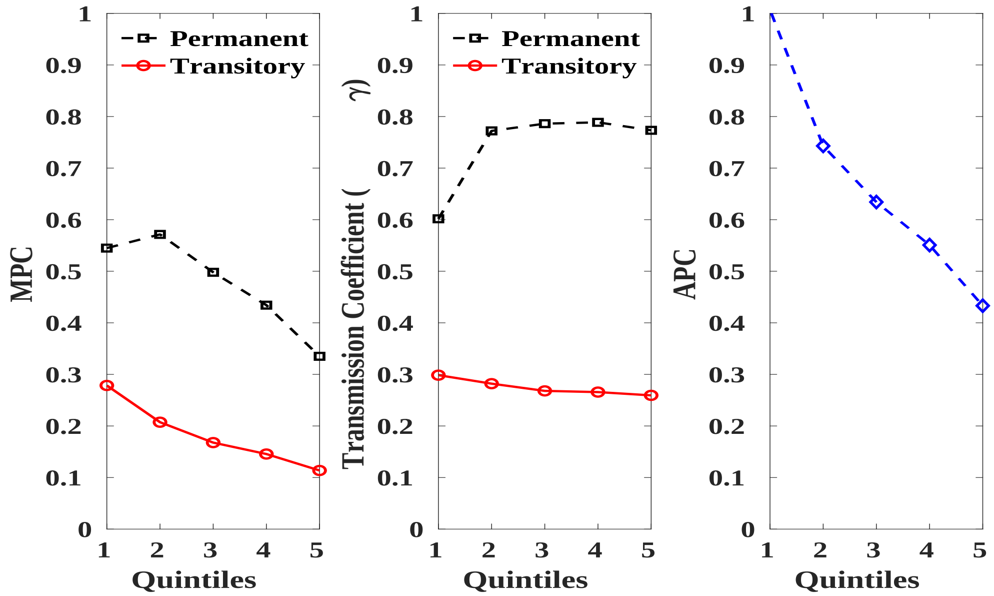
In this section, we use the model to perform several counterfactual exercises to examine the welfare implications of income risk and the role of tax instruments on consumption behavior and welfare.

5.1 Welfare gains from removing income risk

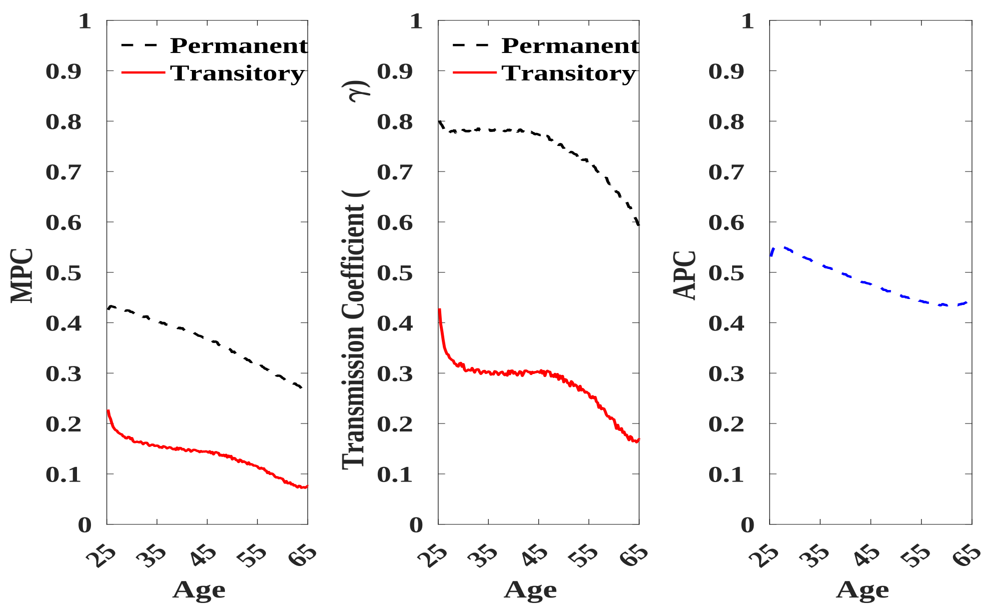
We compute welfare gains from removing idiosyncratic income risk for households as:

$$W = \left[\frac{(1/N)\sum_i V^{\text{cf}}(A,\zeta,\tau,1)}{(1/N)\sum_i V^{*}(A,\zeta,\tau,1)} \right]^{\frac{1}{1-\sigma}} - 1 \tag{5.1}$$

Figure 8: Heterogeneity in Transmission Coefficient, APC and MPC



(a) Lagged Permanent Income



(b) Age

where V^{cf} and V^* are the value function associated at age 1 in the counterfactual and benchmark model, respectively.

Table 5 shows significant welfare gains associated with removing income risk. Household permanent consumption is almost 40% higher in the absence of both temporary and permanent income risk. Welfare increases by 32.79% with removing permanent income risk , while it rises by 8.5% with removing transitory risk. These values are higher than compared to the literature on US like Wu and Krueger (2021) (25%) and Karahan and Ozkan (2013) (16.8%), as these papers estimate a much smaller variance of permanent risk. Moreover, the model is estimated on annual data for Wu and Krueger (2021), while our values are based on quarterly data.

Table 5: Welfare gains from removing income risk

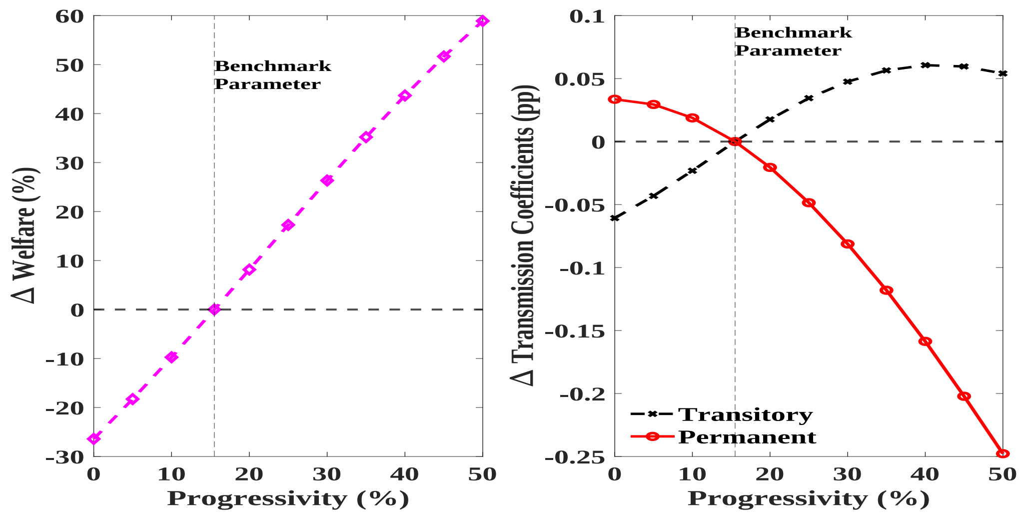
Welfare gain (%)	Value
Removing temporary risk	8.47%
Removing permanent risk	32.79%
Removing all risk	40.16%

5.2 Policy design

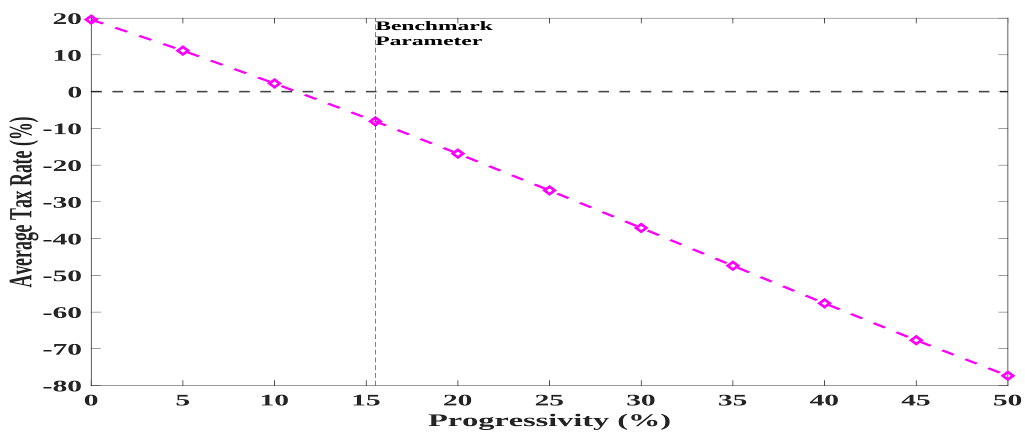
To understand the role of income tax progressivity and consumption tax in impacting consumer welfare and insurance, we conduct counterfactual exercises where we consider different values of a tax parameter. To ensure that the net revenue of the government remains the same as in the benchmark model, we vary another tax instrument. Note, the government holds three tax instruments: ψ, χ_y and χ_c. In each experiment, we will consider specific values of one tax instrument while changing another to maintain net-revenue neutrality. The third tax parameter will remain unchanged during that experiment.

5.2.1 Alternate values of tax progressivity while varying average income tax for neutrality

In this set of counterfactuals, we consider alternate values of the tax progressivity ψ, while we vary χ_y to ensure net revenue neutrality. The consumption tax rate remains at the baseline value across all counterfactuals.



(a) Welfare and Transmission Coefficient

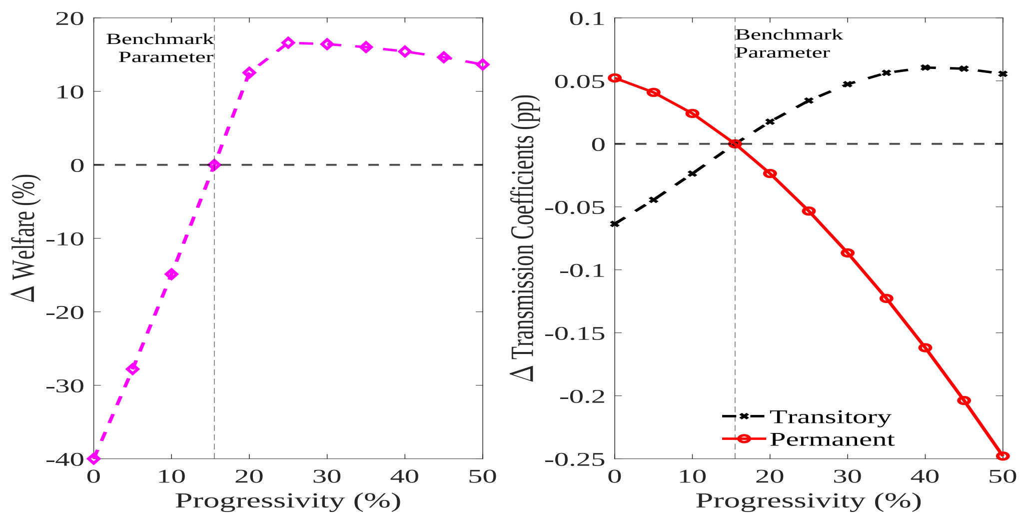


(b) Average Income Tax rate

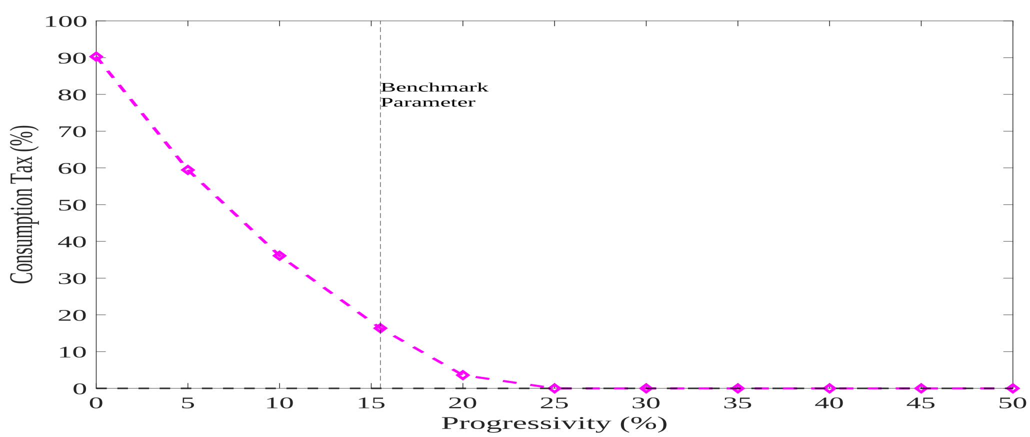
Figure 9a shows that welfare increases with higher tax progressivity. The government subsidizes individuals by reducing the average income tax rate, 1-χ_y. χ_y at baseline is 1.08 implying an average income tax rate of -8%. Higher tax progressivity benefits asset-poor and low-income households significantly by reducing the pass through of a permanent income shock into consumption. The transmission coefficient to a temporary income shock rises as households reduce wealth accumulation and absorb temporary changes of income in consumption. Given the significant income risk in the model, the effect of higher consumption insurance to permanent income shock on welfare dominates, implying that higher tax progressivity increases welfare significantly.

5.2.2 Alternate values of tax progressivity while varying consumption tax for neutrality

We consider alternate values of the tax progressivity ψ, while we vary the consumption tax χ_c to ensure net revenue neutrality. The average income tax rate remains at the baseline value.



(a) Welfare and Transmission Coefficient

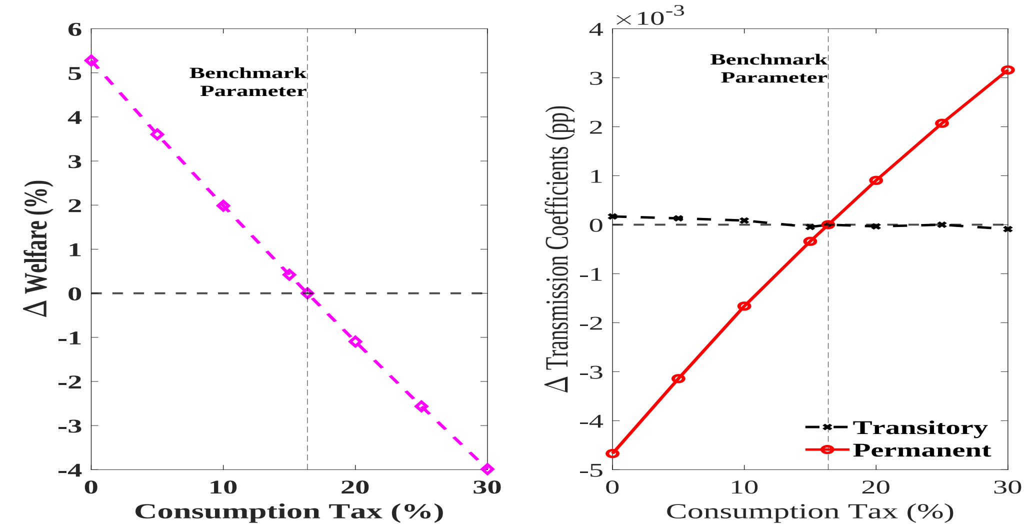


(b) Consumption Tax Rate

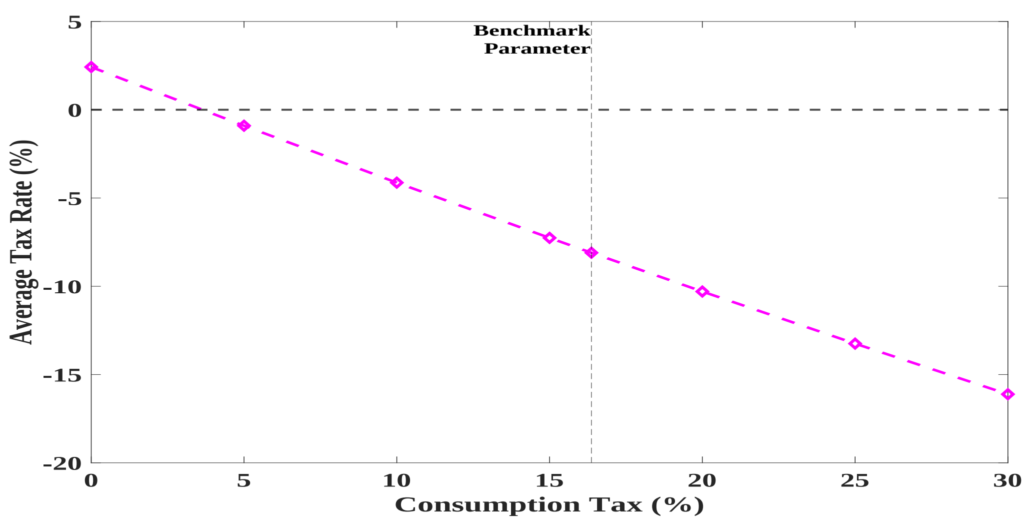
Figure 10a shows that welfare increases with higher tax progressivity, but upto the point that consumption tax become zero. When consumption taxes cannot go negative, while the average income tax also remains fixed, then the government is collecting more revenue without redistributing it back to the households. Thus, welfare gains peak at 25% tax progressivity when consumption tax is zero and the average income tax is -8%. Similar as before, higher tax progressivity reduces the pass through of a permanent income shock into consumption and increases the pass through of a temporary income shock.

5.2.3 Alternate values of consumption tax while varying average income tax for neutrality

We consider alternate values of the consumption tax χ_c, while we vary χ_y to ensure net revenue neutrality.



(a) Welfare and Transmission Coefficient



(b) Average Tax rate

Lowering consumption tax unequivocally improves welfare, as shown in figure 11a. There is little effect on the pass through of a temporary income shock into consumption as the higher average income tax rate offsets the higher consumption tax. However, the pass through of a permanent income shock into consumption is much smaller as consumption tax falls. Note that in this experiment, tax progressivity is at the baseline parameter value of 15.5%. Hence, a higher average tax rate in a progressive tax system instead of a proportional consumption tax, benefits low-income households more. Thus, decline in proportional consumption tax improves consumer welfare and increases consumption insurance against income risk.

The key take-way from these exercises is that optimal policy in an environment with significant background income risk involves high income tax progressivity and zero proportional consumption tax.

6 Discussion

We have an exceptionally rich laboratory in the world's most populated country to study heterogeneity in consumer behavior. The panel data of income and consumption is unique in its richness, both due to its large panel dimension and the higher frequency of observations. Our estimation method leads to precise estimates of consumption insurance, high income risk and variation in both income risk and consumer behavior across households. We show that a standard calibrated life cycle model captures the empirically observed consumption and asset behavior, as well as the consumption response to income fluctuations. Moreover, the framework allows us to conduct policy experiments to show the importance of income tax progressivity and consumption tax rates in a setting with highly volatile income. But there are rich dimensions of heterogeneity that can be explored further to enrich our understanding.

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A Data Appendix

Table A.1: Variables Used and Cleaning Steps

Variable	**Cleaning Steps or Description**
Wage Earnings	Monthly nominal earnings, avg. within HH-quarter
Non-durable consumption	Monthly nominal nondurable consumption, avg. within HH-quarter
Age	Age of first person as listed, min by HH-wave-year
Family members	Count of family members by HH-wave-year
State	State of origin, but state when defining big city
Whether kids	Dummy = 1 if zero kids, at HH-wave-year
Number of kids	Count of HH members with age<18, at HH-wave-year
Religion	As reported
Caste category	As reported
Education	Dummy = 1 if education of first member < 8th standard Dummy = 2 if education of first member b/w 8-12th standard Dummy = 3 if education of first member > 12th standard
Big city	8 districts of Delhi with 1,787 unique HHs Gautam Buddha Nagar with 821 unique HHs Mumbai and its suburban district with 1,777 unique HHs Thane district with 5,077 unique HHs District of Chennai with 801 unique HHs Bangalore urban and rural districts with 1,738 unique HHs Kolkata district with 778 unique HHs Hyderabad district with 1,198 unique HHs Gurgaon district with 592 unique HHs
Other income	Number of earning members other than HoH and spouse
Employment status	Dummy for working
Agri household	Dummy for HoH being employed in agri
Rural	Rural dummy at HH-wave-year
Earnings decile	Decile of the average HH-wave level income b/w 2016-19
Married	Max of whether a spouse exists within HH-wave-year

Notes: This table provides the steps undertaken in constructing each variable used in our analysis. In particular, we explain how the quarterly data on household earnings and consumption, and other variables are constructed.

Table A.2: Sampling Restrictions

Step	**Unique Households**
Raw CMIE people of India data	234,892
Raw CMIE income/consumption data	233,583
Merged POI, income/cons data	231,371
Sample Restrictions	
Drop HH-wave-year with married dummy = 0	195,714
Drop if age < 25 or age > 65	184,863
Keep years between 2016 and 2019	155,268
Trim top and bottom 1% of income growth	154,986
Changed agri HH status b/w 2016 to 2019	154,986
Drop if missing values in > 11 of 16 quarters	90,817
Final Sample	90,817

B State-Space Representation

In this appendix, we present the state-space representation of the unobserved components version of the BPP model.

Suppressing household-specific subscripts for simplicity, the observation equation is

$$\mathbf{y}_t = \mathbf{H}\mathbf{X_t},$$

where

$$\mathbf{y}_t = \begin{bmatrix} y_t \\ c_t \end{bmatrix}, \mathbf{H} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & \gamma_\eta & 1 \end{bmatrix}, \text{ and } \mathbf{X_t} = \begin{bmatrix} y_t - \tau_t \\ c_t - \gamma_\eta \tau_t - \kappa_t \\ \tau_t \\ \kappa_t \end{bmatrix}.$$

The state equation is

$$\mathbf{X}_t = \mathbf{F}\mathbf{X_{t-1}} + \mathbf{v_t},$$

where

$$\mathbf{F} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \ \mathbf{v_t} = \begin{bmatrix} \epsilon_t + \theta\epsilon_{t-1} \\ \upsilon_t \\ \eta_t \\ u_t + \gamma_\epsilon \epsilon_t \end{bmatrix},$$

and the covariance matrix of $\mathbf{v}_t$, $\mathbf{Q}$, is given by

$$\mathbf{Q} = \begin{pmatrix} \sigma_\epsilon^2(1+\theta^2) & 0 & 0 & \gamma_\epsilon\sigma_\epsilon^2 \\ 0 & \sigma_\upsilon^2 & 0 & 0 \\ 0 & 0 & \sigma_\eta^2 & 0 \\ \gamma_\epsilon\sigma_\epsilon^2 & 0 & 0 & \sigma_u^2 + \gamma_\epsilon^2\sigma_\epsilon^2 \end{pmatrix}.$$

Given the state-space representation and an assumption of Normality, we can use the Kalman filter to calculate the likelihood for the BPP model based on the prediction error decomposition and an assumption of independence of idiosyncratic income and consumption across households. In addition, the Kalman filter can be easily adapted to handle missing observations, which are prevalent in any household panel dataset.

C Full Sets of Estimates

Table A.3: Income Risk and Consumption Insurance Estimates

		Full Sample	Primary	Secondary	College
γ_ε		0.278 (0.002)	0.290 (0.003)	0.273 (0.002)	0.266 (0.004)
γ_τ		0.731 (0.004)	0.809 (0.007)	0.723 (0.008)	0.627 (0.009)
σ_u		0.068 (0.001)	0.065 (0.001)	0.068 (0.002)	0.074 (0.002)
σ_ν		0.153 (0.001)	0.153 (0.001)	0.152 (0.001)	0.157 (0.001)
θ		0.295 (0.004)	0.292 (0.005)	0.297 (0.006)	0.301 (0.010)
σ_η	2016Q1	0.317 (0.016)	0.302 (0.040)	0.312 (0.061)	0.363 (0.100)
	Q2	0.117 (0.003)	0.099 (0.007)	0.124 (0.005)	0.147 (0.010)
	Q3	0.230 (0.002)	0.200 (0.003)	0.232 (0.003)	0.293 (0.005)
	Q4	0.209 (0.002)	0.185 (0.003)	0.209 (0.004)	0.264 (0.006)
	2017Q1	0.199 (0.003)	0.174 (0.003)	0.201 (0.003)	0.249 (0.006)
	Q2	0.193 (0.002)	0.173 (0.003)	0.194 (0.004)	0.239 (0.006)
	Q3	0.208 (0.003)	0.187 (0.003)	0.210 (0.004)	0.251 (0.006)
	Q4	0.194 (0.002)	0.176 (0.003)	0.193 (0.003)	0.235 (0.006)
	2018Q1	0.180 (0.002)	0.167 (0.004)	0.176 (0.004)	0.219 (0.006)
	Q2	0.180 (0.002)	0.167 (0.003)	0.179 (0.004)	0.208 (0.006)
	Q3	0.174 (0.002)	0.160 (0.004)	0.177 (0.004)	0.200 (0.006)
	Q4	0.180 (0.002)	0.170 (0.004)	0.180 (0.004)	0.204 (0.006)
	2019Q1	0.172 (0.003)	0.159 (0.004)	0.174 (0.004)	0.197 (0.006)
	Q2	0.180 (0.003)	0.164 (0.004)	0.182 (0.004)	0.207 (0.007)
	Q3	0.179 (0.003)	0.163 (0.004)	0.181 (0.004)	0.210 (0.008)
	Q4	0.261 (0.002)	0.243 (0.003)	0.261 (0.004)	0.293 (0.006)
σ_ϵ	2016Q1	0.210 (0.000)	0.204 (0.001)	0.205 (0.001)	0.235 (0.002)
	Q2	0.216 (0.001)	0.210 (0.002)	0.212 (0.002)	0.235 (0.003)
	Q3	0.215 (0.001)	0.208 (0.002)	0.213 (0.002)	0.233 (0.004)
	Q4	0.221 (0.001)	0.213 (0.002)	0.221 (0.002)	0.236 (0.004)
	2017Q1	0.223 (0.001)	0.218 (0.002)	0.224 (0.002)	0.230 (0.004)
	Q2	0.232 (0.001)	0.225 (0.002)	0.233 (0.002)	0.244 (0.004)
	Q3	0.233 (0.001)	0.230 (0.002)	0.232 (0.002)	0.243 (0.004)
	Q4	0.240 (0.001)	0.237 (0.002)	0.243 (0.002)	0.243 (0.004)
	2018Q1	0.242 (0.001)	0.236 (0.002)	0.244 (0.002)	0.253 (0.004)
	Q2	0.245 (0.001)	0.239 (0.002)	0.249 (0.002)	0.251 (0.004)
	Q3	0.242 (0.002)	0.237 (0.002)	0.242 (0.002)	0.258 (0.004)
	Q4	0.239 (0.001)	0.232 (0.002)	0.241 (0.002)	0.254 (0.004)
	2019Q1	0.237 (0.002)	0.228 (0.002)	0.241 (0.003)	0.252 (0.005)
	Q2	0.240 (0.002)	0.236 (0.003)	0.238 (0.003)	0.257 (0.006)
	Q3	0.207 (0.002)	0.203 (0.003)	0.210 (0.004)	0.218 (0.007)
	Q4	0.153 (0.001)	0.153 (0.001)	0.152 (0.001)	0.157 (0.001)

D Model Appendix

Figure B.1: Life cycle profiles for standard deviation